higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRICAL PRINCIPLES AND PRACTICE
NQF LEVEL 4

18 NOVEMBER 2019

This marking guideline consists of 8 pages.

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QUESTION 1

1.1
- Linear network
- Non-linear network
- Active network
- Passive network (4)

1.2 1.2.1

$$R_{p1}=\frac{R_4\times R_5}{R_4+R_5} \qquad \vee \qquad \frac{1}{R_{P1}}=\frac{1}{R_4}+\frac{1}{R_5}$$

$$=\frac{6\times 3}{6+3} \qquad = \frac{1}{6}+\frac{1}{3}$$

$$=\frac{18}{9} \qquad = \frac{3}{6}$$

$$R_{p1}= \frac{6}{3}=2\ \Omega$$

$$R_S=R_{P1}+R_2+R_6$$

$$=2+4+8$$

$$=14\ \Omega$$

$$R_{p2}=\frac{R_S\times R_3}{R_S+R_3} \qquad \vee \qquad \frac{1}{R_{P2}}=\frac{1}{R_S}+\frac{1}{R_3}$$

$$=\frac{14\times 2}{14+2} \qquad = \frac{1}{14}+\frac{1}{2}$$

$$=\frac{28}{16} \qquad =\frac{8}{14}$$

$$R_{P2}= \frac{14}{8}=1{,}75\,\Omega$$

$$R_T=R_{P2}+R_1$$

$$=1{,}75+2$$

$$=3{,}75\ \Omega$$ (7)

1.2.2

$$I_{R1}=I_T=\frac{V_T}{R_T}$$

$$=\frac{12}{3{,}75}$$

$$=3{,}2\ A$$ (2)

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1.2.3 $R_T = R_1 + R_3$

$= 2 + 2$

$= 4\ \Omega$ (2)

1.2.4 $I_T = \frac{V_T}{R_T}$

$= \frac{12}{4}$

$= 3\ A$ (2)

1.3 1.3.1 $V_{rms} = 0,707\, V_m$

$= 0,707 \times 200$

$= 141,4\ V$ (2)

1.3.2 $\omega = 2\pi f$

$f = \frac{\omega}{2\pi}$

$= \frac{628}{2\pi}$

$= 99,949 \quad \vee \quad 100\ Hz$ (2)

1.3.3 $v = 200\ \sin\ 2\pi ft$

$= 200\ \sin\ (2\pi \times 100 \times 0,002)$

$= 200\ \sin\ 1,2566$

$= 190,209\ V$

OR

$v = 200\ \sin\ 2\pi ft$

$= 200\ \sin\ (2 \times 180^0 \times 100 \times 0,002)$

$= 200\ \sin\ 72^0$

$= 190,211\ V$ (3)

1.4 The form factor is the ratio of the rms value to the average value,✓ whereas the peak factor is the ratio of the peak or maximum value to the rms value.✓ (2)

1.5 $cos\theta = 0,6$

$\theta = \cos^{-1} 0,6$

$= 53,13^0$

For $Q = S \sin\theta \ \ (note: S = VI)$

$= 7000 \times \sin\ 53,13^0$✓

$= 5599,99\ \ Var \quad \vee \quad 5,6\ \ kVar$ (3)

1.6 The maximum value is the maximum instantaneous value measured from its zero value or the largest value reached in a half cycle.✓ (1)

[30]

QUESTION 2

2.1 In a Hydro-electric power station, water head is created by constructing a dam across a river or lake. From the dam, water is led to a water turbine✓.
The water turbine captures the energy in the falling water and changes the hydraulic energy into mechanical energy at the turbine shaft✓. The turbine drives the alternator which converts mechanical energy into electrical energy✓. (3)

2.2
- It is cheaper to generate three-phase current than to generate single-phase current.
- A three-phase distribution system can also provide single-phase current.
- Three-phase motors need less maintenance.
- A rotating magnetic field is easily produced in a three-phase induction motor.
- There is less power loss in the value for a given amount of load power.
- Three-phase motors produce a constant shaft torque, resulting in less variation and vibration.
- Rectified three-phase does not require as much filtering as rectified single-phase.
- Three-phase systems need less copper or less conducting material than single-phase systems for a given voltage rating.
- The power factor of single-phase motors is poor compared to three-phase motors of the same rating. (Any 6 × 1) (6)

2.3 In star connection, line voltage is root three times phase voltage (VL = √3 VP),✓ whereas in delta connection, the line voltage is equal to phase voltage (VL = VP).✓ (2)

2.4 Poles are used to lead conductors from one place to another. (1)

2.5 $P = \sqrt{3} V_L I_L \cos \theta$

$= \sqrt{3} \times 380 \times 12 \times 0{,}9$

$= 7108{,}337 \ W$

$= 7{,}108 \ kW$ (3)

[15]

QUESTION 3

3.1 3.1.1

- To maintain the potential of any part of a system at a definite value with respect to earth,
- To allow fault current to flow to earth in the event of a fault, so that the protective gear starts operating to isolate the fault current.
- To ensure that in the event of a fault, the apparatus cannot reach dangerous potential with respect to earth

(3)

3.1.2

- A trench earth
- An earth rod
- A foundation earth (3)

3.2

- Switch off the main switch
- Use a multimeter or a low-resistance ohmmeter
- Connect one end of the test lead to the consumer's earth terminal and other end to all the conductive metal parts of the installation one at a time
- The reading must not be greater than 0.2 Ω

(Any 3 x1) (3)

3.3 One or more conductive parts that are embedded in the earth to make effective electrical contact with general mass of earth . (2)

3.4 Earth leakage current (1)

3.5 3.5.1 Part of the system uses a combined protective earth and neutral (PEN) conductor from the transformer to the building's distribution point.✓Separate protective earth (PE) and neutral (N) conductors are used in the building's fixed indoor wiring and flexible power cords.✓ (2)

3.5.2 Protective earth and neutral are separate conductors that are connected together only near the power source.

(1)

[15]

QUESTION 4

4.1

- The primary winding will have thinner conductor while the secondary winding conductor will be thicker
- Primary windings will have more turns, while secondary windings will be lesser
- When measuring the resistance across the two terminals, the primary winding must have higher resistance since it is formed by winding many turns of thinner copper wire than the secondary (resistance increases with an increase in length and decreases with the decrease in length), while the secondary side will have lower resistance. (3)

4.2 $Full\ \ load\ \ output\ \ power = VI\cos\ \theta$

$$= 300 \times 10^3 \times 0{,}85$$

$$= 255\ \ kW$$

$$P_{IN} = P_{OUT} + losses$$

$$= 1{,}9 + 255$$

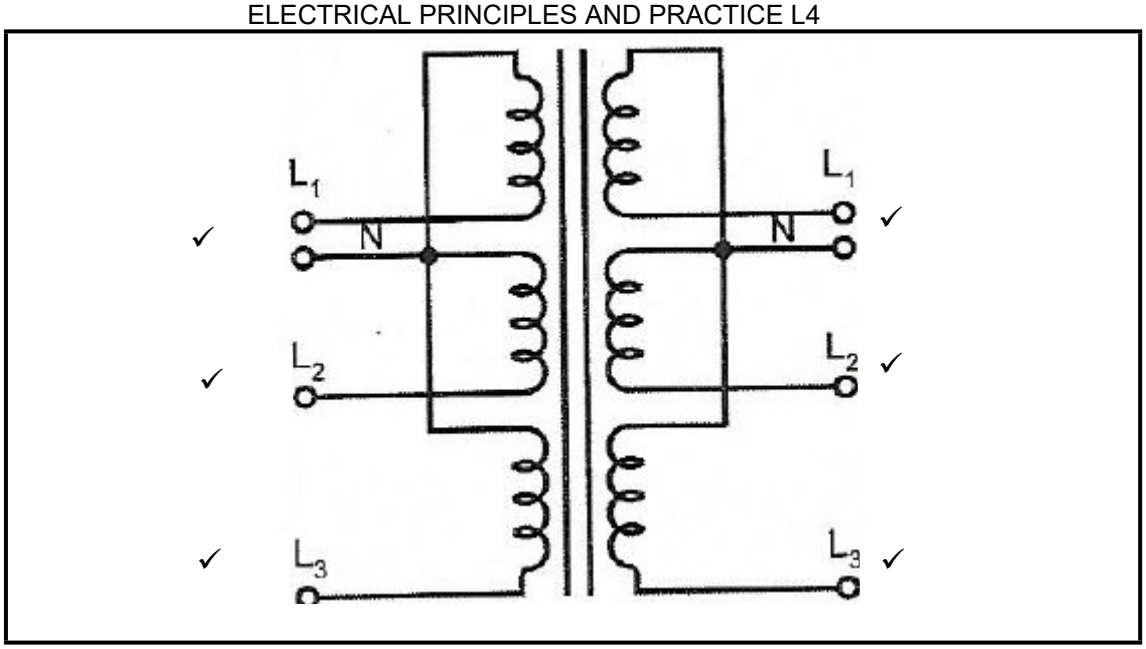
$$= 256{,}9\ \ kW$$

$$Efficiency\ \ (\eta) = \frac{P_{IN} - losses}{P_{IN}}$$

$$Efficiency\ \ (\eta) = \frac{256{,}9 - 1{,}9}{256{,}9}$$

$$= 0{,}99\ \vee\ 99\,\%$$ (5)

4.3



(6)

4.4 The hollow core transformer is shaped with a hallow square through the center , while the shell type has layers consisting of E and I shaped section of meta

OR

With the hollow core windings surrounds a considerable part of the core , whereas in the shell type transformer the core surrounds a considerable portion of the windings. (2)

4.5 4.5.1 A delta-delta connection is suitable for large LV transformers.

4.5.2 A star-star is used in some industrial facilities where large currents need to be carried on low voltages.

4.5.3 A delta-star connection can be used in most industrial and commercial applications.

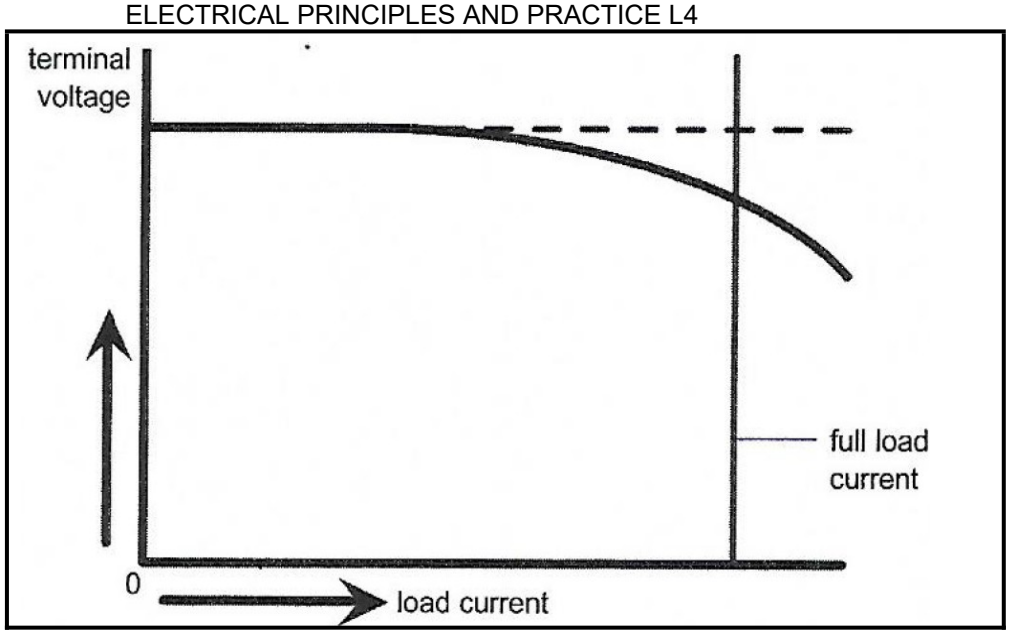
4.5.4 Star-delta connections are used in HV transmission.

(4 × 1) (4)

[20]

QUESTION 5

5.1 5.1.1



Correct sketch = 1 mark
Correct labels = 3 marks (4)

5.1.2 The shape is nearly straight at the beginning✓ because flux produced is proportional to the exciting current;✓ but after a certain point, the shape starts to curve downwards✓ due to voltage drop caused by the armature reaction and resistance drop due to circuit resistance.✓ (4)

5.2 When a three-phase stator windings of an induction motor are fed by a three-phase supply, a magnetic flux of constant magnitude, but rotating at synchronous speed is set up.✓ The flux passes through the air-gap, sweeps past the rotor surface and so cuts the rotor conductors which as yet, are stationary.✓ Due to the relative speed between the rotating flux and stationery rotor conductors, e.m.f. is induced in the lattter and thus current starts flowing through the rotor conductors.✓ Consequently, mechanical force acts on the rotor conductors and produces a torque which tends to move the rotor in the same direction as the rotating field.✓ (4)

5.3 5.3.1

$$N_s = \frac{60\,f}{p} \qquad \vee \qquad n_s = \frac{f}{p}$$

$$= \frac{60 \times 60}{3} \qquad = \frac{60}{3}$$

$$= 1200 \ \ rev/min \qquad = 20 \ \ r/s$$

(2)

5.3.2

$$s = \frac{N_s - N_r}{N_s} \times 100\,\%$$

$$= \frac{1200 - 900}{1200} \times 100\,\%$$

(2)

$= 25\ \%$

OR

$$s = \frac{n_s - n_r}{n_s} \times 100\,\%$$

$$= \frac{20 - 15}{20} \times 100\,\%$$

$= 25\ \%$

5.4 The rotation direction of a compound motor can be reversed by reversing the connections of either the field windings or the armature , but not both. (2)

5.5
- The purpose of open-circuit test is to determine no-load loss or core loss and no load current
- To determine the efficiency of the motor(no load power factor)
- To synchronise the speed of the motor (Any 2 × 1) (2)

[20]

TOTAL: 100